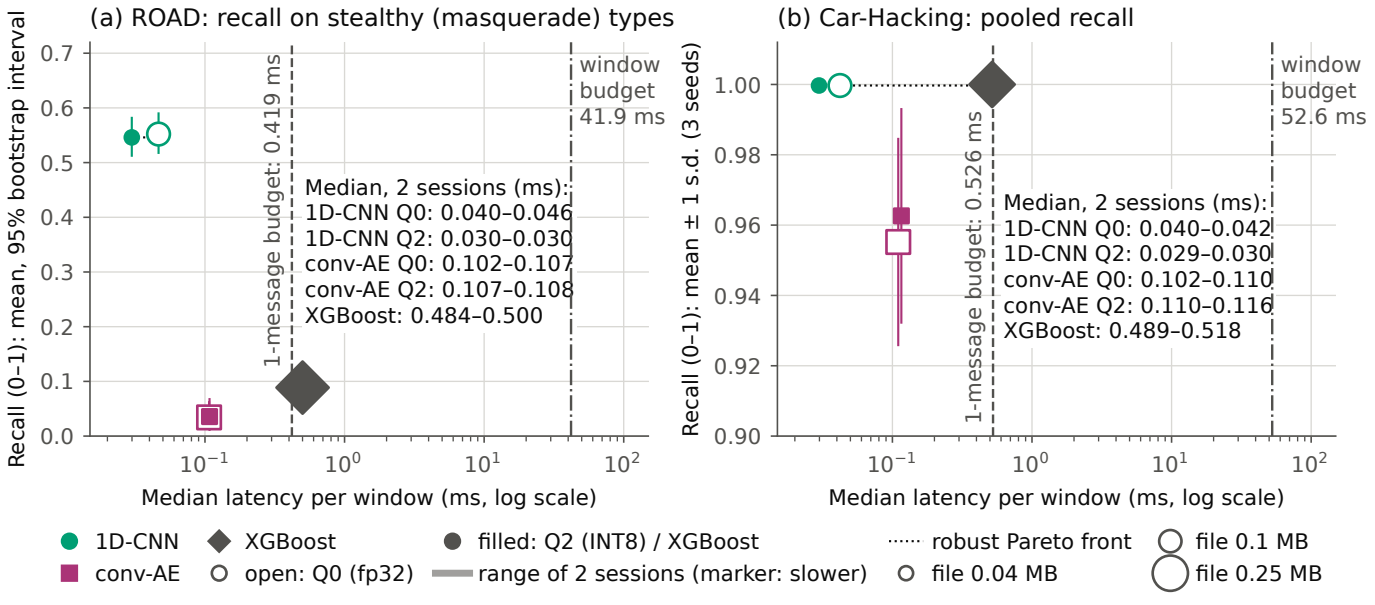
(a) ROAD: recall on stealthy (masquerade) types
Recall (0–1): mean, 95% bootstrap interval
1-message budget: 0.419 ms
window budget 41.9 ms
Median, 2 sessions (ms):
1D-CNN Q0: 0.040–0.046
1D-CNN Q2: 0.030–0.030
conv-AE Q0: 0.102–0.107
conv-AE Q2: 0.107–0.108
XGBoost: 0.484–0.500
Median latency per window (ms, log scale)
(b) Car-Hacking: pooled recall
Recall (0–1): mean ± 1 s.d. (3 seeds)
1-message budget: 0.526 ms
window budget 52.6 ms
Median, 2 sessions (ms):
1D-CNN Q0: 0.040–0.042
1D-CNN Q2: 0.029–0.030
conv-AE Q0: 0.102–0.110
conv-AE Q2: 0.110–0.116
XGBoost: 0.489–0.518
Median latency per window (ms, log scale)
1D-CNN
XGBoost
filled: Q2 (INT8) / XGBoost
robust Pareto front
file 0.1 MB
conv-AE
open: Q0 (fp32)
range of 2 sessions (marker: slower)
file 0.04 MB
file 0.25 MB